NGHI NGUYEN

nghi.28.nguyen@gmail.com • www.linkedin.com/in/nghinguyen1 • https://nghi-nguyen.netlify.app/

EDUCATION

Bachelor of Science in Computer Science and Engineering — **September 2019 - May 2023**
Massachusetts Institute of Technology (MIT) — *GPA: 3.7 / 4.0*

WORK EXPERIENCE

AI Engineer — **January 2026 – Present**
International Business Machines (IBM), Data + AI-First Transformation — *New York, NY*

- Designed an LLM agent evaluation framework (**Langfuse**, **AgentOps**), defining production-readiness metrics and gold-set benchmarks across 3 business domains, securing Chief Data Officer investment in continued evaluation development
- Established a move-to-production framework for LLM agent deployment encompassing evaluation gates, human-in-the-loop tieouts, and QA checkpoints, reducing bug recovery time by 90% post-release
- Led semantic model architecture redesign as tech lead, aligning 6 engineers, data scientists, and platform users through standups, design reviews, and engineering syncs to deliver within a 2-week timeline alongside the existing system
- Delivered a production AI insight-agent platform on **watsonx Orchestrate** in 90 days, driving adoption among 300+ users
- Identified scalability limitations on the current platform architecture, securing department buy-in for a skills-based architecture that eliminated domain misrouting and improved tool response time by 60%

Data Engineer II / Data Engineer — **August 2023 – January 2026**
International Business Machines (IBM), Chief Data Office — *New York, NY*

- Owned and scaled 20+ ETL pipelines on **Kubernetes**, processing 180M+ records using **Python**, **PySpark**, **SQL**, and **Apache Iceberg**, reducing delivery delays by 70%
- Partnered cross-functionally with data SMEs, product managers, and stakeholders to align on requirements early, eliminating rework cycles and reducing scope creep, cutting average revision rounds by 75% per project
- Designed and deployed a scalable data ingestion framework for departmental productivity metrics using Jira, Kubernetes, **Spark**, Apache Iceberg, **Docker** and **AWS S3**, ultimately adopted by 6 engineering teams to surface bottlenecks and optimize agile workflows
- Built automated data validation and quality checks across 30+ pipelines, catching failures before downstream execution and cutting issue-resolution time by 80%
- Leveraged engineering expertise to prototype and deliver pipeline architecture for new reporting capabilities, eliminating handoffs and cutting project delivery timelines by 50%
- Owned on-call support for pipelines serving 30K+ users, ensuring SLA compliance through rapid incident response

RESEARCH EXPERIENCE

Research Assistant — **September 2021 – January 2023**
MIT CSAIL App Inventor – Prof. Hal Abelson — *Cambridge, MA*

- Ran zero-shot to multi-shot prompt experiments on GPT-3 to evaluate code-generation quality for MIT App Inventor; found multi-shot prompting produced more consistent, valid code than zero-shot, with returns diminishing beyond a certain number of examples
- Analyzed cross-cultural survey data (WEIRD/non-WEIRD) on trust in conversational agents; findings co-authored into two peer-reviewed papers (AAAI 2023, ACM IDC 2023)
- Presented research on trust in conversational agents at Brave Conversations Barcelona 2022

PUBLICATIONS

Teaching Conversational AI for Building Appropriate Trust — AAAI 2023

- Co-led educational workshops with children and parents on conversational AI; analyzed survey and qualitative data to examine changes in agent perceptions and trust, and co-developed recommendations for teaching conversational AI concepts and fostering appropriate trust

How Children and Parents Perceive AI Agents — ACM IDC 2023

- Helped design and conduct a user study with children and parents examining perceptions, trust, and partner models of conversational agents; analyzed survey and qualitative data and co-authored recommendations for transparent, trustworthy agent design

SKILLS AND INTERESTS

Skills

- LLM Evaluation: agent evaluation and observability, benchmark design, gold-set curation, human-in-the-loop review, Langfuse, AgentOps
- Research: experiment design, survey and qualitative analysis, human-subjects research, prompt experiments
- Engineering: Python, SQL, PySpark, Spark, Iceberg, Kubernetes, Docker, Airflow, AWS S3

Additional Information

- Professional Interests: AI Safety and Governance, HCI, LLM Evaluations, Education Accessibility, Data Quality
- Languages: French (conversational), Vietnamese (conversational), Japanese (beginner)